Zihao Zhang

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PROFILE

I received my M.S. degree in the School of Computer Science at Fudan University, advised by Prof. Yu-Gang Jiang and Prof. Zuxuan Wu.

My research explores generative visual modeling for **embodied intelligence**, with the goal of building **controllable and efficient visual foundations** that help embodied agents understand and predict how scenes evolve over time.

My work has been published at leading computer vision and multimedia venues, including **CVPR**, **ICCV**, **ACM MM** and **SIGGRAPH Asia**.

RESEARCH INTERESTS

- **Generative world models** and controllable video generation for embodied intelligence
- **Spatial reasoning**, camera planning, and motion understanding
- **Efficient diffusion models** for high-resolution visual generation

EDUCATION

Sep 2023 — Jun 2026 **Fudan University (QS 26)**
Master's Degree · Shanghai, China
School of Computer Science · GPA: 3.5 · Advisors: Prof. Yu-Gang Jiang & Prof. Zuxuan Wu

Sep 2018 — Jun 2022 **Wuhan University (QS 165)**
Bachelor's Degree · Wuhan, China
School of Chemistry and Molecular Sciences · GPA: 3.3

AWARDS & HONORS

2025 **First-Class Academic Scholarship** · Fudan University

2020 & 2021 **First-Class Scholarship and Grant** · Wuhan University

2025 **EDEN Patent Application** · Huawei Noah's Ark Lab · Large-motion video frame interpolation

RESEARCH EXPERIENCE

Apr 2025 — Jun 2026 Research Intern · **BiliBili Inc.** — Shanghai, China

- **Temporal Dynamics Modeling**
  SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation— *First author, ACM MM 2026*
  - Model continuous visual dynamics from surrounding observations to infer high-fidelity intermediate states efficiently.
  - Developed a one-step pixel diffusion framework with progressive coarse-to-fine dynamics modeling, improving both fidelity and inference speed.

  MinVideo: Minute-scale High-quality Video Generation— *Ongoing project*
  - Model long-horizon visual dynamics to sustain high-quality video generation over minute-scale sequences.
  - Developed recurrent segment conditioning with identity anchors, preserving temporal continuity and subject consistency throughout long-form generation.
- **Spatial Reasoning and Action Prediction**
  CT-1: Vision-Language-Camera Models for Camera-Controllable Video Generation— *Co-first author, ACM MM 2026*
  - Ground camera-controllable video generation in spatial reasoning from visual observations and language instructions.
  - Developed a Vision-Language-Camera model for coherent viewpoint planning and spatially consistent scene evolution.

  FM-WAM: Fast World Action Model via Mask— *Ongoing project*
  - Focus future-action prediction on spatial information that is relevant to an agent's behavior.
  - Developed mask-guided visual modeling that filters irrelevant context, improving prediction efficiency and reliability.

Jul 2024 — Mar 2025 Research Intern · **Huawei Noah's Ark Lab** — Advised by Hang Xu · Shanghai, China

- **Large-motion Temporal Modeling**
  EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation— *First author, CVPR 2025*
  - Model complex temporal transitions to infer realistic intermediate states under large and nonlinear scene motion.
  - Developed a motion-aware diffusion framework that preserves structural details and temporal coherence during high-quality frame interpolation.

- **Controllable Human Motion Editing**
  MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion— *Third author, ICCV 2025*
  - Enable controllable human-motion editing while preserving subject identity, scene appearance, and original camera dynamics.
  - Contributed to a lightweight motion-guided diffusion framework for expressive motion transfer in complex and dynamic environments.

Publications

01 **EDEN: Enhanced Diffusion for High-quality Large-motion Video Frame Interpolation**
Zihao Zhang, Haoran Chen, Haoyu Zhao, Guansong Lu, Yanwei Fu, Hang Xu, Zuxuan Wu
CVPR 2025

02 **SPEED: One-Step Pixel Diffusion for High-quality Video Frame Interpolation**
Zihao Zhang*, Haoyu Zhao*, Siqian Yang, Yidi Wu, Yudong Jiang, Zuxuan Wu
ACM MM 2026

03 **CT-1: Vision-Language-Camera Models Transfer Spatial Reasoning Knowledge to Camera-Controllable Video Generation**
Haoyu Zhao*, **Zihao Zhang***, Jiaxi Gu, Haoran Chen, Qingping Zheng, Pin Tang, Yeying Jin, Yuang Zhang, Junqi Cheng, Zenghui Lu, Peng Shu, Zuxuan Wu, Yu-Gang Jiang
ACM MM 2026

04 **MotionFollower: Editing Video Motion via Lightweight Score-Guided Diffusion**
Shuyuan Tu, Qi Dai, **Zihao Zhang**, Sicheng Xie, Zhi-Qi Cheng, Chong Luo, Xintong Han, Zuxuan Wu, Yu-Gang Jiang
ICCV 2025

05 **VIDiff: Translating Videos via Multi-Modal Instructions with Diffusion Models**
Zhen Xing, Qi Dai, **Zihao Zhang**, Hui Zhang, Han Hu, Zuxuan Wu, Yu-Gang Jiang
arXiv:2311.18837, 2023

06 **AniME: Adaptive Multi-Agent Planning for Long Animation Generation**
Lisai Zhang, Baohan Xu, Siqian Yang, Mingyu Yin, Jing Liu, Chao Xu, Siqi Wang, Yidi Wu, Yuxin Hong, **Zihao Zhang**, Yanzhang Liang, Yudong Jiang
SIGGRAPH Asia 2025 Posters

07 **EVA: Enhancing Anime Video Generation via Reinforcement Learning**
Yidi Wu, Bingwen Zhu, **Zihao Zhang**, Siqian Yang, Lisai Zhang, Baohan Xu, Mingyu Yin, Yanzhang Liang, Yudong Jiang
ICASSP 2026

RESEARCH AGENDA

- **Memory-Persistent Interactive World Models**
  - Build persistent memory mechanisms that maintain coherent environment states across long-horizon interactions.
  - Develop efficient generation for real-time prediction of how visual environments evolve under agent actions.
- **Action-Video Consistent World Action Models**
  - Explore the bidirectional relationship between agent actions and resulting visual dynamics.
  - Explore mutual verification between action-conditioned video prediction and video-grounded action inference for closed-loop self-correction.